Encaminhamento de Dados

-- Trabalho Pratico 2025/2026 --

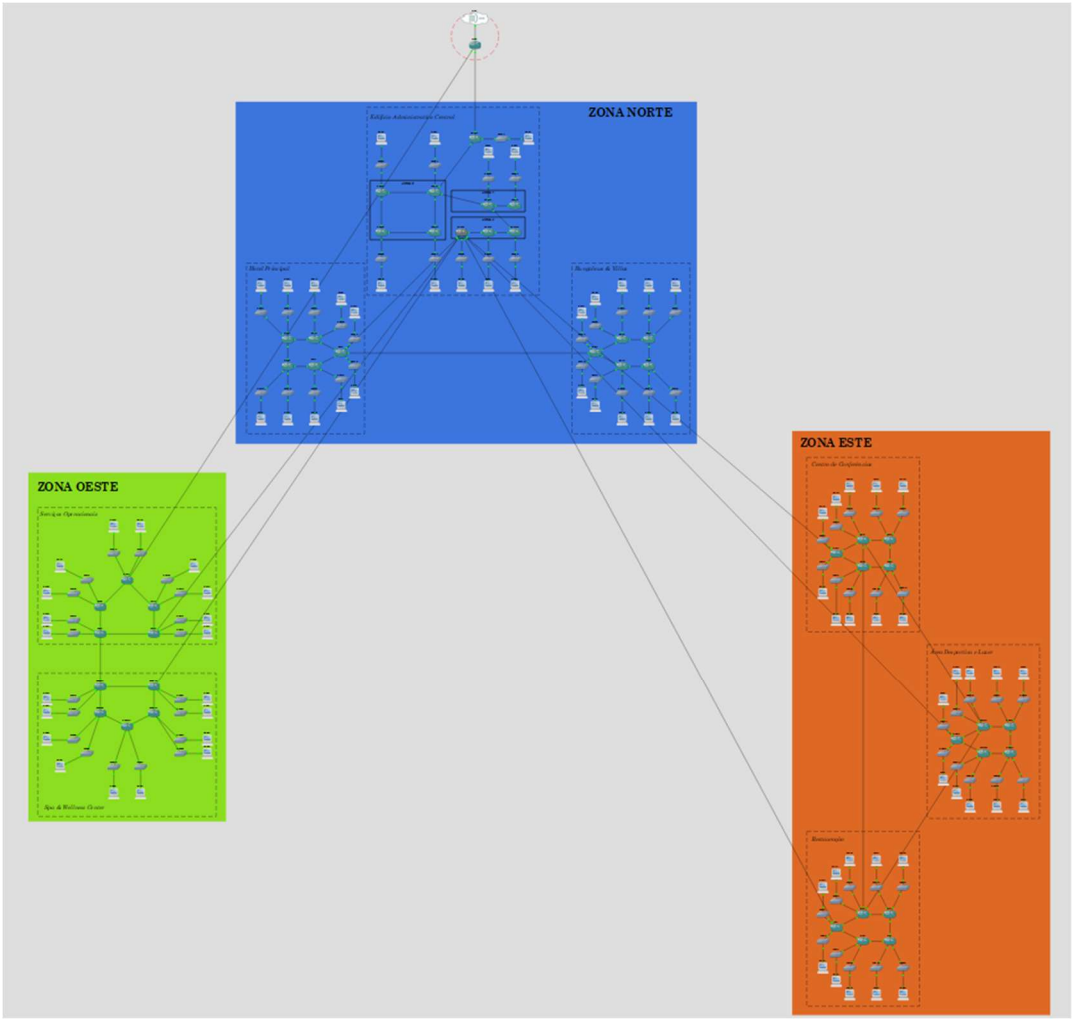


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Licenciatura em Engenharia Informática

Ramo de Redes e Administração de Sistemas

- *Topologia do Trabalho:*



- *Introdução:*

Este trabalho pratico foi feito a partir do GNS3 v2.2.58.

Os routers utilizados foram

Os switch utilizados foram

Todos os routers tem configurado uma password (cisco), igual em todos.

- *Protocolos de Encaminhamento usados em cada setor:*

  Neste trabalho prático foram utilizados 3 protocolos de encaminhamento para interligar os três setores do Atlantic Bay Resort & SPA. O protocolo OSPF foi implementado no Edifício Administrativo Central, Spa & Wellness e Restauração. O protocolo EIGRP foi configurado no Hotel Principal, Serviços Operacionais e Centro de Conferências. Por fim, o protocolo RIP (version 2) foi utilizado nas áreas de Bungalows & Villas e na Área Desportiva e Lazer.

  Em todos os routers foi configurada uma palavra-passe de acesso global, um *banner motd* de identificação do resort e a sincronização de logs (logging synchronous). Para garantir a segurança das atualizações de routing, foi colocada autenticação MD5 em todas as vizinhanças OSPF e autenticação por *key-chain* nas redes RIP. Além disso, foi garantido que as sumarizações automáticas estavam desativadas nos protocolos *Distance Vector* para um encaminhamento preciso (classless).

  Devido à natureza multiprotocolo da rede, routers de fronteira (como o SO1 e o CC1) foram configurados como ASBR (*Autonomous System Boundary Routers*). Nestes equipamentos foi efetuada a redistribuição mútua entre protocolos (ex: redistribute eigrp no OSPF e vice-versa), permitindo a comunicação ponta-a-ponta entre todos os edifícios.

  Para garantir que todos os setores têm acesso à Internet, foi implementado NAT Overload (PAT) nos routers de saída. Para tal, foram criadas ACLs (Access Control Lists) abrangentes que permitem mascarar todos os endereços privados das várias sub-redes da topologia, garantindo o sucesso do tráfego externo (como comprovado nos testes de *ping* ao 2.2.2.2).

  No Setor Oeste, foi implementada Engenharia de Tráfego (Traffic Engineering). Utilizando offset-lists no protocolo EIGRP, procedeu-se à manipulação artificial de métricas, obrigando o tráfego a preferir o router SO1 como caminho primário e deixando o SO3 como via de contingência (redundância).

  Adicionalmente, na zona do Centro de Conferências (routers CC1 e CC2), foi implementado o endereçamento IPv6. Esta implementação incluiu a configuração de roteamento IPv6 e a criação de túneis dinâmicos (6to4) para encapsular e transportar o tráfego IPv6 através da infraestrutura IPv4 existente.

- *Tabelas de Encaminhamento*

EAC-1:

```
EAC1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.20.97.249 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.20.97.249
      10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C        10.0.0.0/30 is directly connected, Ethernet0/1
L        10.0.0.1/32 is directly connected, Ethernet0/1
O        10.0.0.4/30 [110/20] via 10.0.0.2, 00:44:42, Ethernet0/1
O        10.0.0.8/30 [110/50] via 10.0.0.2, 00:44:42, Ethernet0/1
O        10.0.0.12/30 [110/30] via 10.0.0.2, 00:44:42, Ethernet0/1
O        10.0.0.16/30 [110/40] via 10.0.0.2, 00:44:42, Ethernet0/1
O        10.0.0.20/30 [110/20] via 10.0.0.2, 00:44:42, Ethernet0/1
O IA     10.0.0.24/30 [110/30] via 10.0.0.2, 00:44:42, Ethernet0/1
O IA     10.0.0.32/30 [110/30] via 10.0.0.2, 00:44:42, Ethernet0/1
O IA     10.0.0.36/30 [110/40] via 10.0.0.2, 00:44:42, Ethernet0/1
C        10.20.97.248/30 is directly connected, Ethernet0/0
L        10.20.97.250/32 is directly connected, Ethernet0/0
S     194.65.0.0/16 [1/0] via 10.0.0.2
      194.65.98.0/24 is variably subnetted, 9 subnets, 2 masks
C        194.65.98.0/29 is directly connected, Ethernet0/2
L        194.65.98.1/32 is directly connected, Ethernet0/2
O        194.65.98.8/29 [110/20] via 10.0.0.2, 00:44:42, Ethernet0/1
O        194.65.98.16/29 [110/30] via 10.0.0.2, 00:44:42, Ethernet0/1
O        194.65.98.24/29 [110/40] via 10.0.0.2, 00:44:42, Ethernet0/1
O        194.65.98.32/29 [110/50] via 10.0.0.2, 00:44:42, Ethernet0/1
O IA     194.65.98.40/29 [110/30] via 10.0.0.2, 00:44:42, Ethernet0/1
O IA     194.65.98.48/29 [110/40] via 10.0.0.2, 00:44:42, Ethernet0/1
O IA     194.65.98.72/29 [110/40] via 10.0.0.2, 00:44:42, Ethernet0/1
```

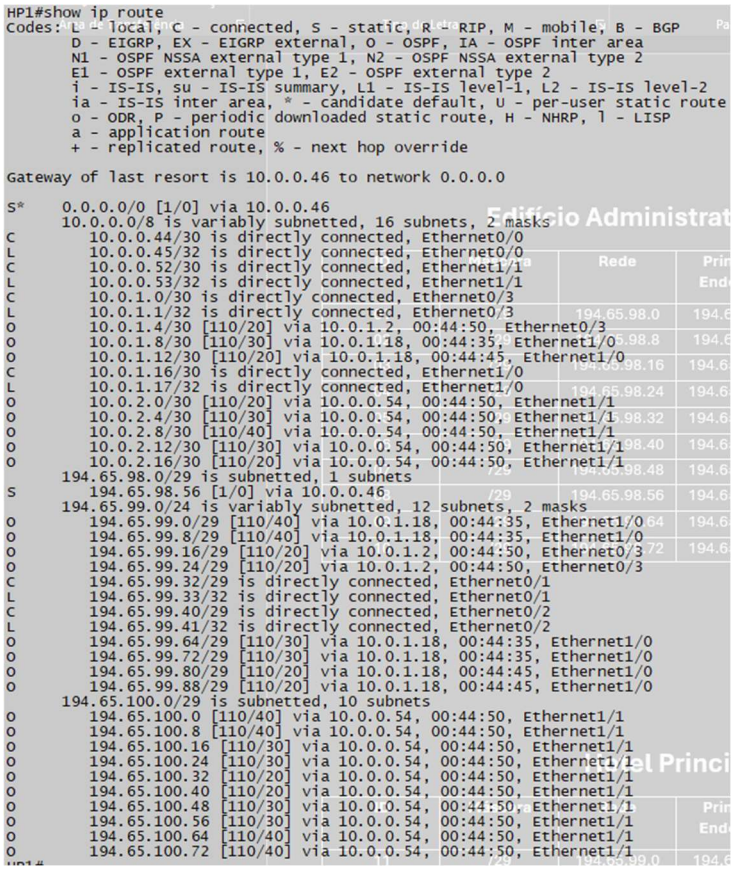
EAC-8:

```
EAC8#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS l
       ia - IS-IS inter area, * - candidate default, U - per-user stat
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LIS
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.42 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.0.42
      10.0.0.0/8 is variably subnetted, 37 subnets, 3 masks
O        10.0.0.36/30 [110/20] via 10.0.0.42, 00:44:47, Ethernet0/0
C        10.0.0.40/30 is directly connected, Ethernet0/0
L        10.0.0.41/32 is directly connected, Ethernet0/0
C        10.0.0.44/30 is directly connected, Ethernet0/2
L        10.0.0.46/32 is directly connected, Ethernet0/2
C        10.0.0.56/30 is directly connected, Ethernet0/3
L        10.0.0.58/32 is directly connected, Ethernet0/3
C        10.0.0.60/30 is directly connected, Ethernet1/0
L        10.0.0.61/32 is directly connected, Ethernet1/0
C        10.0.0.64/30 is directly connected, Ethernet1/1
L        10.0.0.65/32 is directly connected, Ethernet1/1
C        10.0.0.68/30 is directly connected, Ethernet1/2
L        10.0.0.69/32 is directly connected, Ethernet1/2
C        10.0.0.72/30 is directly connected, Ethernet1/3
L        10.0.0.73/32 is directly connected, Ethernet1/3
O E2     10.0.3.0/30 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     10.0.3.4/30 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     10.0.3.8/30 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     10.0.3.12/30 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     10.0.3.16/30 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O        10.0.4.0/30 [110/20] via 10.0.0.62, 00:44:57, Ethernet1/0
O        10.0.4.4/30 [110/20] via 10.0.0.62, 00:45:07, Ethernet1/0
O        10.0.4.8/30 [110/30] via 10.0.0.62, 00:44:57, Ethernet1/0
O        10.0.4.12/30 [110/40] via 10.0.0.62, 00:44:57, Ethernet1/0
O        10.0.4.16/30 [110/30] via 10.0.0.62, 00:44:57, Ethernet1/0
S        10.0.5.0/24 [1/0] via 10.0.0.66
O E2     10.0.5.0/30 [110/20] via 10.0.0.66, 00:45:49, Ethernet1/1
O E2     10.0.5.4/30 [110/20] via 10.0.0.66, 00:45:49, Ethernet1/1
O E2     10.0.5.8/30 [110/20] via 10.0.0.66, 00:45:49, Ethernet1/1
O E2     10.0.6.0/30 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     10.0.6.4/30 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     10.0.6.8/30 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     10.0.6.12/30 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     10.0.6.16/30 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
S        10.0.7.0/24 [1/0] via 10.0.0.74
O E2     10.3.3.12/30 [110/20] via 10.0.0.66, 00:45:49, Ethernet1/1
O E2     10.3.3.16/30 [110/20] via 10.0.0.66, 00:45:49, Ethernet1/1
      194.65.98.0/24 is variably subnetted, 3 subnets, 2 masks
C        194.65.98.56/29 is directly connected, Ethernet0/1
L        194.65.98.57/32 is directly connected, Ethernet0/1
O        194.65.98.64/29 [110/20] via 10.0.0.42, 00:44:47, Ethernet0/0
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     194.65.101.8/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     194.65.101.16/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     194.65.101.24/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     194.65.101.32/29 [110/20] via 10.0.0.57, 00:44:47, Ethernet0/3
O E2     194.65.101.40/29 [110/20] via 10.0.0.57, 00:44:47, Ethernet0/3
O E2     194.65.101.48/28 [110/20] via 10.0.0.57, 00:44:47, Ethernet0/3
O E2     194.65.101.48/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     194.65.101.56/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     194.65.101.64/28 [110/20] via 10.0.0.57, 00:44:47, Ethernet0/3
O E2     194.65.101.64/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
O E2     194.65.101.72/29 [110/20] via 10.0.0.57, 00:45:44, Ethernet0/3
      194.65.103.0/24 is variably subnetted, 11 subnets, 2 masks
S        194.65.103.0/24 [1/0] via 10.0.0.66
O        194.65.103.0/29 [110/20] via 10.0.0.66, 00:45:49, Ethernet1/1
                         [110/20] via 10.0.0.62, 00:45:39, Ethernet1/0
O        194.65.103.8/29 [110/20] via 10.0.0.66, 00:45:49, Ethernet1/1
                         [110/20] via 10.0.0.62, 00:45:39, Ethernet1/0
O        194.65.103.16/29 [110/30] via 10.0.0.62, 00:44:57, Ethernet1/0
O        194.65.103.24/29 [110/30] via 10.0.0.62, 00:44:57, Ethernet1/0
O        194.65.103.32/29 [110/40] via 10.0.0.62, 00:44:57, Ethernet1/0
O        194.65.103.40/29 [110/40] via 10.0.0.62, 00:44:57, Ethernet1/0
O        194.65.103.48/29 [110/40] via 10.0.0.62, 00:44:57, Ethernet1/0
O        194.65.103.56/29 [110/40] via 10.0.0.62, 00:44:57, Ethernet1/0
O        194.65.103.64/29 [110/30] via 10.0.0.62, 00:44:57, Ethernet1/0
O        194.65.103.72/29 [110/30] via 10.0.0.62, 00:44:57, Ethernet1/0
      194.65.104.0/29 is subnetted, 10 subnets
O        194.65.104.0 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O        194.65.104.8 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     194.65.104.16 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     194.65.104.24 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     194.65.104.32 [110/20] via 10.0.0.70, 00:45:28, Ethernet1/2
O E2     194.65.104.40 [110/20] via 10.0.0.70, 00:45:27, Ethernet1/2
O E2     194.65.104.48 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     194.65.104.56 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     194.65.104.64 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
O E2     194.65.104.72 [110/20] via 10.0.0.70, 00:45:39, Ethernet1/2
      194.65.105.0/24 is variably subnetted, 3 subnets, 2 masks
S        194.65.105.0/24 [1/0] via 10.0.0.74
O E2     194.65.105.0/29 [110/20] via 10.0.0.74, 00:45:49, Ethernet1/3
O E2     194.65.105.8/29 [110/20] via 10.0.0.74, 00:45:49, Ethernet1/3
```

HP-1:

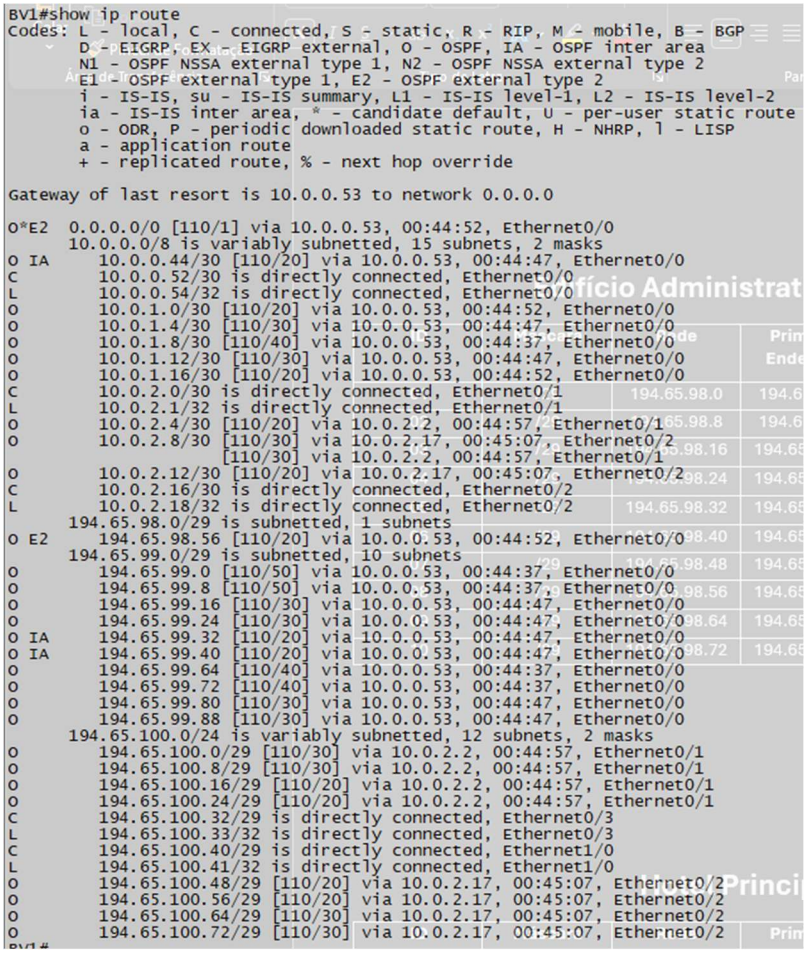


```
HP1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.46 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.0.46
      10.0.0.0/8 is variably subnetted, 16 subnets, 2 masks
C        10.0.0.44/30 is directly connected, Ethernet0/0
L        10.0.0.45/32 is directly connected, Ethernet0/0
C        10.0.0.52/30 is directly connected, Ethernet1/1
L        10.0.0.53/32 is directly connected, Ethernet1/1
C        10.0.1.0/30 is directly connected, Ethernet0/3
L        10.0.1.1/32 is directly connected, Ethernet0/3
O        10.0.1.4/30 [110/20] via 10.0.1.2, 00:44:50, Ethernet0/3
O        10.0.1.8/30 [110/30] via 10.0.1.18, 00:44:35, Ethernet1/0
O        10.0.1.12/30 [110/20] via 10.0.1.18, 00:44:45, Ethernet1/0
C        10.0.1.16/30 is directly connected, Ethernet1/0
L        10.0.1.17/32 is directly connected, Ethernet1/0
O        10.0.2.0/30 [110/20] via 10.0.0.54, 00:44:50, Ethernet1/1
O        10.0.2.4/30 [110/30] via 10.0.0.54, 00:44:50, Ethernet1/1
O        10.0.2.8/30 [110/40] via 10.0.0.54, 00:44:50, Ethernet1/1
O        10.0.2.12/30 [110/30] via 10.0.0.54, 00:44:50, Ethernet1/1
O        10.0.2.16/30 [110/20] via 10.0.0.54, 00:44:50, Ethernet1/1
      194.65.98.0/29 is subnetted, 1 subnets
S        194.65.98.56 [1/0] via 10.0.0.46
      194.65.99.0/24 is variably subnetted, 12 subnets, 2 masks
O        194.65.99.0/29 [110/40] via 10.0.1.18, 00:44:35, Ethernet1/0
O        194.65.99.8/29 [110/40] via 10.0.1.18, 00:44:35, Ethernet1/0
O        194.65.99.16/29 [110/20] via 10.0.1.2, 00:44:50, Ethernet0/3
O        194.65.99.24/29 [110/20] via 10.0.1.2, 00:44:50, Ethernet0/3
C        194.65.99.32/29 is directly connected, Ethernet0/1
L        194.65.99.33/32 is directly connected, Ethernet0/1
C        194.65.99.40/29 is directly connected, Ethernet0/2
L        194.65.99.41/32 is directly connected, Ethernet0/2
O        194.65.99.64/29 [110/30] via 10.0.1.18, 00:44:35, Ethernet1/0
O        194.65.99.72/29 [110/30] via 10.0.1.18, 00:44:35, Ethernet1/0
O        194.65.99.80/29 [110/20] via 10.0.1.18, 00:44:45, Ethernet1/0
O        194.65.99.88/29 [110/20] via 10.0.1.18, 00:44:45, Ethernet1/0
      194.65.100.0/29 is subnetted, 10 subnets
O        194.65.100.0 [110/40] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.8 [110/40] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.16 [110/30] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.24 [110/30] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.32 [110/20] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.40 [110/20] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.48 [110/30] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.56 [110/30] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.64 [110/40] via 10.0.0.54, 00:44:50, Ethernet1/1
O        194.65.100.72 [110/40] via 10.0.0.54, 00:44:50, Ethernet1/1
```

BV-1:



```
BV1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.53 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.0.0.53, 00:44:52, Ethernet0/0
      10.0.0.0/8 is variably subnetted, 15 subnets, 2 masks
O IA     10.0.0.44/30 [110/20] via 10.0.0.53, 00:44:47, Ethernet0/0
C        10.0.0.52/30 is directly connected, Ethernet0/0
L        10.0.0.54/32 is directly connected, Ethernet0/0
O        10.0.1.0/30 [110/20] via 10.0.0.53, 00:44:52, Ethernet0/0
O        10.0.1.4/30 [110/30] via 10.0.0.53, 00:44:47, Ethernet0/0
O        10.0.1.8/30 [110/40] via 10.0.0.53, 00:44:37, Ethernet0/0
O        10.0.1.12/30 [110/30] via 10.0.0.53, 00:44:47, Ethernet0/0
O        10.0.1.16/30 [110/20] via 10.0.0.53, 00:44:52, Ethernet0/0
C        10.0.2.0/30 is directly connected, Ethernet0/1
L        10.0.2.1/32 is directly connected, Ethernet0/1
O        10.0.2.4/30 [110/20] via 10.0.2.2, 00:44:57, Ethernet0/1
O        10.0.2.8/30 [110/30] via 10.0.2.17, 00:45:07, Ethernet0/2
                     [110/30] via 10.0.2.2, 00:44:57, Ethernet0/1
O        10.0.2.12/30 [110/20] via 10.0.2.17, 00:45:07, Ethernet0/2
C        10.0.2.16/30 is directly connected, Ethernet0/2
L        10.0.2.18/32 is directly connected, Ethernet0/2
      194.65.98.0/29 is subnetted, 1 subnets
O E2     194.65.98.56 [110/20] via 10.0.0.53, 00:44:52, Ethernet0/0
      194.65.99.0/29 is subnetted, 10 subnets
O        194.65.99.0 [110/50] via 10.0.0.53, 00:44:37, Ethernet0/0
O        194.65.99.8 [110/50] via 10.0.0.53, 00:44:37, Ethernet0/0
O        194.65.99.16 [110/30] via 10.0.0.53, 00:44:47, Ethernet0/0
O        194.65.99.24 [110/30] via 10.0.0.53, 00:44:47, Ethernet0/0
O IA     194.65.99.32 [110/20] via 10.0.0.53, 00:44:47, Ethernet0/0
O IA     194.65.99.40 [110/20] via 10.0.0.53, 00:44:47, Ethernet0/0
O        194.65.99.64 [110/40] via 10.0.0.53, 00:44:37, Ethernet0/0
O        194.65.99.72 [110/40] via 10.0.0.53, 00:44:37, Ethernet0/0
O        194.65.99.80 [110/30] via 10.0.0.53, 00:44:47, Ethernet0/0
O        194.65.99.88 [110/30] via 10.0.0.53, 00:44:47, Ethernet0/0
      194.65.100.0/24 is variably subnetted, 12 subnets, 2 masks
O        194.65.100.0/29 [110/30] via 10.0.2.2, 00:44:57, Ethernet0/1
O        194.65.100.8/29 [110/30] via 10.0.2.2, 00:44:57, Ethernet0/1
O        194.65.100.16/29 [110/20] via 10.0.2.2, 00:44:57, Ethernet0/1
O        194.65.100.24/29 [110/20] via 10.0.2.2, 00:44:57, Ethernet0/1
C        194.65.100.32/29 is directly connected, Ethernet0/3
L        194.65.100.33/32 is directly connected, Ethernet0/3
C        194.65.100.40/29 is directly connected, Ethernet1/0
L        194.65.100.41/32 is directly connected, Ethernet1/0
O        194.65.100.48/29 [110/20] via 10.0.2.17, 00:45:07, Ethernet0/2
O        194.65.100.56/29 [110/20] via 10.0.2.17, 00:45:07, Ethernet0/2
O        194.65.100.64/29 [110/30] via 10.0.2.17, 00:45:07, Ethernet0/2
O        194.65.100.72/29 [110/30] via 10.0.2.17, 00:45:07, Ethernet0/2
```

SO-1:

```
SO1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static rou
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.58 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.0.58
      10.0.0.0/8 is variably subnetted, 32 subnets, 3 masks
O        10.0.0.36/30 [110/30] via 10.0.0.58, 00:44:47, Ethernet0/3
O        10.0.0.40/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O        10.0.0.44/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
C        10.0.0.56/30 is directly connected, Ethernet0/3
L        10.0.0.57/32 is directly connected, Ethernet0/3
O        10.0.0.60/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O        10.0.0.64/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O        10.0.0.68/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O        10.0.0.72/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
C        10.0.3.0/30 is directly connected, Ethernet0/0
L        10.0.3.1/32 is directly connected, Ethernet0/0
D        10.0.3.4/30 [90/307200] via 10.0.3.2, 00:45:53, Ethernet0/0
D        10.0.3.8/30 [90/332800] via 10.0.3.17, 00:44:52, Ethernet1/0
                     [90/332800] via 10.0.3.2, 00:44:52, Ethernet0/0
D        10.0.3.12/30 [90/307200] via 10.0.3.17, 00:45:53, Ethernet1/0
C        10.0.3.16/30 is directly connected, Ethernet1/0
L        10.0.3.18/32 is directly connected, Ethernet1/0
O        10.0.4.0/30 [110/30] via 10.0.0.58, 00:44:57, Ethernet0/3
O        10.0.4.4/30 [110/30] via 10.0.0.58, 00:45:07, Ethernet0/3
O        10.0.4.8/30 [110/40] via 10.0.0.58, 00:44:57, Ethernet0/3
O        10.0.4.12/30 [110/50] via 10.0.0.58, 00:44:57, Ethernet0/3
O        10.0.4.16/30 [110/40] via 10.0.0.58, 00:44:57, Ethernet0/3
O E2     10.0.5.0/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O E2     10.0.5.4/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O E2     10.0.5.8/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O E2     10.0.6.0/30 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     10.0.6.4/30 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     10.0.6.8/30 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     10.0.6.12/30 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     10.0.6.16/30 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O IA     10.0.7.0/24 [110/30] via 10.0.0.58, 00:45:40, Ethernet0/3
O E2     10.3.3.12/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O E2     10.3.3.16/30 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O        194.65.98.64 [110/30] via 10.0.0.58, 00:44:47, Ethernet0/3
      194.65.101.0/24 is variably subnetted, 14 subnets, 3 masks
C        194.65.101.0/29 is directly connected, Ethernet0/1
L        194.65.101.1/32 is directly connected, Ethernet0/1
C        194.65.101.8/29 is directly connected, Ethernet0/2
L        194.65.101.9/32 is directly connected, Ethernet0/2
D        194.65.101.16/29 [90/307200] via 10.0.3.2, 00:45:53, Ethernet0/0
D        194.65.101.24/29 [90/307200] via 10.0.3.2, 00:45:53, Ethernet0/0
D        194.65.101.32/29 [90/332800] via 10.0.3.2, 00:44:52, Ethernet0/0
D        194.65.101.40/29 [90/332800] via 10.0.3.2, 00:44:52, Ethernet0/0
D EX     194.65.101.48/28 [170/332800] via 10.0.3.2, 00:44:52, Ethernet0/0
D        194.65.101.48/29 [90/332800] via 10.0.3.17, 00:44:47, Ethernet1/0
D        194.65.101.56/29 [90/332800] via 10.0.3.17, 00:44:47, Ethernet1/0
D EX     194.65.101.64/28 [170/332800] via 10.0.3.2, 00:44:52, Ethernet0/0
D        194.65.101.64/29 [90/307200] via 10.0.3.17, 00:45:53, Ethernet1/0
D        194.65.101.72/29 [90/307200] via 10.0.3.17, 00:45:53, Ethernet1/0
      194.65.103.0/29 is subnetted, 10 subnets
O        194.65.103.0 [110/30] via 10.0.0.58, 00:45:40, Ethernet0/3
O        194.65.103.8 [110/30] via 10.0.0.58, 00:45:40, Ethernet0/3
O        194.65.103.16 [110/40] via 10.0.0.58, 00:44:57, Ethernet0/3
O        194.65.103.24 [110/40] via 10.0.0.58, 00:44:57, Ethernet0/3
O        194.65.103.32 [110/50] via 10.0.0.58, 00:44:57, Ethernet0/3
O        194.65.103.40 [110/50] via 10.0.0.58, 00:44:57, Ethernet0/3
O        194.65.103.48 [110/50] via 10.0.0.58, 00:44:57, Ethernet0/3
O        194.65.103.56 [110/50] via 10.0.0.58, 00:44:57, Ethernet0/3
O        194.65.103.64 [110/40] via 10.0.0.58, 00:44:57, Ethernet0/3
O        194.65.103.72 [110/40] via 10.0.0.58, 00:44:57, Ethernet0/3
      194.65.104.0/29 is subnetted, 10 subnets
O        194.65.104.0 [110/30] via 10.0.0.58, 00:45:35, Ethernet0/3
O        194.65.104.8 [110/30] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     194.65.104.16 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     194.65.104.24 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     194.65.104.32 [110/20] via 10.0.0.58, 00:45:27, Ethernet0/3
O E2     194.65.104.40 [110/20] via 10.0.0.58, 00:45:27, Ethernet0/3
O E2     194.65.104.48 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     194.65.104.56 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     194.65.104.64 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
O E2     194.65.104.72 [110/20] via 10.0.0.58, 00:45:35, Ethernet0/3
      194.65.105.0/24 is variably subnetted, 3 subnets, 2 masks
O IA     194.65.105.0/24 [110/40] via 10.0.0.58, 00:45:40, Ethernet0/3
O E2     194.65.105.0/29 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
O E2     194.65.105.8/29 [110/20] via 10.0.0.58, 00:45:40, Ethernet0/3
```

```
SWC1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.61 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.0.0.61, 00:45:38, Ethernet0/3
      10.0.0.0/8 is variably subnetted, 32 subnets, 3 masks
O        10.0.0.36/30 [110/30] via 10.0.0.61, 00:44:47, Ethernet0/3
O        10.0.0.40/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O        10.0.0.44/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O        10.0.0.56/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
C        10.0.0.60/30 is directly connected, Ethernet0/3
L        10.0.0.62/32 is directly connected, Ethernet0/3
O        10.0.0.64/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O        10.0.0.68/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O        10.0.0.72/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.3.0/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.3.4/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.3.8/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.3.12/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.3.16/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
C        10.0.4.0/30 is directly connected, Ethernet0/0
L        10.0.4.1/32 is directly connected, Ethernet0/0
C        10.0.4.4/30 is directly connected, Ethernet1/0
L        10.0.4.5/32 is directly connected, Ethernet1/0
O        10.0.4.8/30 [110/20] via 10.0.4.2, 00:44:57, Ethernet0/0
O        10.0.4.12/30 [110/30] via 10.0.4.6, 00:45:07, Ethernet1/0
                      [110/30] via 10.0.4.2, 00:44:57, Ethernet0/0
O        10.0.4.16/30 [110/20] via 10.0.4.6, 00:45:07, Ethernet1/0
O E2     10.0.5.0/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.5.4/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.5.8/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.6.0/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.6.4/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.6.8/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.6.12/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.0.6.16/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O IA     10.0.7.0/24 [110/30] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.3.3.12/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     10.3.3.16/30 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O        194.65.98.64 [110/30] via 10.0.0.61, 00:44:47, Ethernet0/3
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.101.8/29 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.101.16/29 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.101.24/29 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.101.32/29 [110/20] via 10.0.0.61, 00:44:47, Ethernet0/3
O E2     194.65.101.40/29 [110/20] via 10.0.0.61, 00:44:47, Ethernet0/3
O E2     194.65.101.48/28 [110/20] via 10.0.0.61, 00:44:47, Ethernet0/3
      194.65.103.0/24 is variably subnetted, 12 subnets, 2 masks
C        194.65.103.0/29 is directly connected, Ethernet0/1
L        194.65.103.1/32 is directly connected, Ethernet0/1
C        194.65.103.8/29 is directly connected, Ethernet0/2
L        194.65.103.9/32 is directly connected, Ethernet0/2
O        194.65.103.16/29 [110/20] via 10.0.4.2, 00:44:57, Ethernet0/0
O        194.65.103.24/29 [110/20] via 10.0.4.2, 00:44:57, Ethernet0/0
O        194.65.103.32/29 [110/30] via 10.0.4.2, 00:44:57, Ethernet0/0
O        194.65.103.40/29 [110/30] via 10.0.4.2, 00:44:57, Ethernet0/0
O        194.65.103.48/29 [110/30] via 10.0.4.6, 00:45:07, Ethernet1/0
O        194.65.103.56/29 [110/30] via 10.0.4.6, 00:45:07, Ethernet1/0
O        194.65.103.64/29 [110/20] via 10.0.4.6, 00:45:07, Ethernet1/0
O        194.65.103.72/29 [110/20] via 10.0.4.6, 00:45:07, Ethernet1/0
      194.65.104.0/29 is subnetted, 10 subnets
O        194.65.104.0 [110/30] via 10.0.0.61, 00:45:38, Ethernet0/3
O        194.65.104.8 [110/30] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.104.16 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.104.24 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.104.32 [110/20] via 10.0.0.61, 00:45:27, Ethernet0/3
O E2     194.65.104.40 [110/20] via 10.0.0.61, 00:45:27, Ethernet0/3
O E2     194.65.104.48 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.104.56 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.104.64 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.104.72 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
      194.65.105.0/24 is variably subnetted, 3 subnets, 2 masks
O IA     194.65.105.0/24 [110/40] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.105.0/29 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
O E2     194.65.105.8/29 [110/20] via 10.0.0.61, 00:45:38, Ethernet0/3
```

CC-1:

```
CC1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.65 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.0.65
      10.0.0.0/8 is variably subnetted, 32 subnets, 3 masks
O        10.0.0.36/30 [110/30] via 10.0.0.65, 00:44:47, Ethernet1/0
O        10.0.0.40/30 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        10.0.0.44/30 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        10.0.0.56/30 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        10.0.0.60/30 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
C        10.0.0.64/30 is directly connected, Ethernet1/0
L        10.0.0.66/32 is directly connected, Ethernet1/0
O        10.0.0.68/30 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        10.0.0.72/30 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O E2     10.0.3.0/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.3.4/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.3.8/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.3.12/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.3.16/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O        10.0.4.0/30 [110/30] via 10.0.0.65, 00:44:57, Ethernet1/0
O        10.0.4.4/30 [110/30] via 10.0.0.65, 00:45:07, Ethernet1/0
O        10.0.4.8/30 [110/40] via 10.0.0.65, 00:44:57, Ethernet1/0
O        10.0.4.12/30 [110/50] via 10.0.0.65, 00:44:57, Ethernet1/0
O        10.0.4.16/30 [110/40] via 10.0.0.65, 00:44:57, Ethernet1/0
C        10.0.5.0/30 is directly connected, Ethernet0/0
L        10.0.5.1/32 is directly connected, Ethernet0/0
C        10.0.5.4/30 is directly connected, Ethernet0/1
L        10.0.5.5/32 is directly connected, Ethernet0/1
R        10.0.5.8/30 [120/1] via 10.0.5.2, 00:00:12, Ethernet0/0
O E2     10.0.6.0/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.6.4/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.6.8/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.6.12/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     10.0.6.16/30 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O IA     10.0.7.0/24 [110/30] via 10.0.0.65, 00:45:35, Ethernet1/0
R        10.3.3.12/30 [120/2] via 10.0.5.6, 00:00:08, Ethernet0/1
                      [120/2] via 10.0.5.2, 00:00:12, Ethernet0/0
R        10.3.3.16/30 [120/1] via 10.0.5.6, 00:00:08, Ethernet0/1
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        194.65.98.64 [110/30] via 10.0.0.65, 00:44:47, Ethernet1/0
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.8/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.16/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.24/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.32/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.40/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/28 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.56/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O IA     10.0.7.0/24 [110/30] via 10.0.0.65, 00:45:35, Ethernet1/0
R        10.3.3.12/30 [120/2] via 10.0.5.6, 00:00:08, Ethernet0/1
                      [120/2] via 10.0.5.2, 00:00:12, Ethernet0/0
R        10.3.3.16/30 [120/1] via 10.0.5.6, 00:00:08, Ethernet0/1
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        194.65.98.64 [110/30] via 10.0.0.65, 00:44:47, Ethernet1/0
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.8/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.16/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.24/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.32/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.40/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/28 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.56/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O IA     10.0.7.0/24 [110/30] via 10.0.0.65, 00:45:35, Ethernet1/0
R        10.3.3.12/30 [120/2] via 10.0.5.6, 00:00:08, Ethernet0/1
                      [120/2] via 10.0.5.2, 00:00:12, Ethernet0/0
R        10.3.3.16/30 [120/1] via 10.0.5.6, 00:00:08, Ethernet0/1
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        194.65.98.64 [110/30] via 10.0.0.65, 00:44:47, Ethernet1/0
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.8/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.16/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.24/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.32/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.40/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/28 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.56/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
```

```
O IA     10.0.7.0/24 [110/30] via 10.0.0.65, 00:45:35, Ethernet1/0
R        10.3.3.12/30 [120/2] via 10.0.5.6, 00:00:08, Ethernet0/1
                      [120/2] via 10.0.5.2, 00:00:12, Ethernet0/0
R        10.3.3.16/30 [120/1] via 10.0.5.6, 00:00:08, Ethernet0/1
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O        194.65.98.64 [110/30] via 10.0.0.65, 00:44:47, Ethernet1/0
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.8/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.16/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.24/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.32/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.40/29 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/28 [110/20] via 10.0.0.65, 00:44:47, Ethernet1/0
O E2     194.65.101.48/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.56/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.64/28 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.64/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.101.72/29 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
      194.65.103.0/24 is variably subnetted, 12 subnets, 2 masks
C        194.65.103.0/29 is directly connected, Ethernet0/2
L        194.65.103.1/32 is directly connected, Ethernet0/2
C        194.65.103.8/29 is directly connected, Ethernet0/3
L        194.65.103.9/32 is directly connected, Ethernet0/3
O        194.65.103.16/29 [110/40] via 10.0.0.65, 00:44:57, Ethernet1/0
O        194.65.103.24/29 [110/40] via 10.0.0.65, 00:44:57, Ethernet1/0
S        194.65.103.32/29 [1/0] via 10.0.5.2
S        194.65.103.40/29 [1/0] via 10.0.5.2
O        194.65.103.48/29 [110/50] via 10.0.0.65, 00:44:57, Ethernet1/0
O        194.65.103.56/29 [110/50] via 10.0.0.65, 00:44:57, Ethernet1/0
O        194.65.103.64/29 [110/40] via 10.0.0.65, 00:44:57, Ethernet1/0
O        194.65.103.72/29 [110/40] via 10.0.0.65, 00:44:57, Ethernet1/0
      194.65.104.0/29 is subnetted, 10 subnets
O        194.65.104.0 [110/30] via 10.0.0.65, 00:45:35, Ethernet1/0
O        194.65.104.8 [110/30] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.104.16 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.104.24 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.104.32 [110/20] via 10.0.0.65, 00:45:27, Ethernet1/0
O E2     194.65.104.40 [110/20] via 10.0.0.65, 00:45:27, Ethernet1/0
O E2     194.65.104.48 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.104.56 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.104.64 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.104.72 [110/20] via 10.0.0.65, 00:45:35, Ethernet1/0
      194.65.105.0/24 is variably subnetted, 3 subnets, 2 masks
O IA     194.65.105.0/24 [110/40] via 10.0.0.65, 00:45:35, Ethernet1/0
O E2     194.65.105.0/29 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
O E2     194.65.105.8/29 [110/20] via 10.0.0.65, 00:45:50, Ethernet1/0
```

ADL-1:

```
ADL1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.69 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.0.69
      10.0.0.0/8 is variably subnetted, 32 subnets, 3 masks
O        10.0.0.36/30 [110/30] via 10.0.0.69, 00:44:47, Ethernet1/0
O        10.0.0.40/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O        10.0.0.44/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O        10.0.0.56/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O        10.0.0.60/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O        10.0.0.64/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
C        10.0.0.68/30 is directly connected, Ethernet1/0
L        10.0.0.70/32 is directly connected, Ethernet1/0
O        10.0.0.72/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.0.3.0/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.0.3.4/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.0.3.8/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.0.3.12/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.0.3.16/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O        10.0.4.0/30 [110/30] via 10.0.0.69, 00:44:57, Ethernet1/0
O        10.0.4.4/30 [110/30] via 10.0.0.69, 00:45:07, Ethernet1/0
O        10.0.4.8/30 [110/40] via 10.0.0.69, 00:44:57, Ethernet1/0
O        10.0.4.12/30 [110/50] via 10.0.0.69, 00:44:57, Ethernet1/0
O        10.0.4.16/30 [110/40] via 10.0.0.69, 00:44:57, Ethernet1/0
O E2     10.0.5.0/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.0.5.4/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.0.5.8/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
C        10.0.6.0/30 is directly connected, Ethernet0/0
L        10.0.6.1/32 is directly connected, Ethernet0/0
R        10.0.6.4/30 [120/1] via 10.0.6.2, 00:00:24, Ethernet0/0
R        10.0.6.8/30 [120/2] via 10.0.6.17, 00:00:13, Ethernet0/1
                     [120/2] via 10.0.6.2, 00:00:24, Ethernet0/0
R        10.0.6.12/30 [120/1] via 10.0.6.17, 00:00:13, Ethernet0/1
C        10.0.6.16/30 is directly connected, Ethernet0/1
L        10.0.6.18/32 is directly connected, Ethernet0/1
O IA     10.0.7.0/24 [110/30] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.3.3.12/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.3.3.16/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O        194.65.98.64 [110/30] via 10.0.0.69, 00:44:47, Ethernet1/0
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.8/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.16/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.24/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.32/29 [110/20] via 10.0.0.69, 00:44:47, Ethernet1/0
O E2     194.65.101.40/29 [110/20] via 10.0.0.69, 00:44:47, Ethernet1/0
```

```
C        10.0.6.0/30 is directly connected, Ethernet0/0
L        10.0.6.1/32 is directly connected, Ethernet0/0
R        10.0.6.4/30 [120/1] via 10.0.6.2, 00:00:24, Ethernet0/0
R        10.0.6.8/30 [120/2] via 10.0.6.17, 00:00:13, Ethernet0/1
                     [120/2] via 10.0.6.2, 00:00:24, Ethernet0/0
R        10.0.6.12/30 [120/1] via 10.0.6.17, 00:00:13, Ethernet0/1
C        10.0.6.16/30 is directly connected, Ethernet0/1
L        10.0.6.18/32 is directly connected, Ethernet0/1
O IA     10.0.7.0/24 [110/30] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.3.3.12/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     10.3.3.16/30 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
      194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O        194.65.98.64 [110/30] via 10.0.0.69, 00:44:47, Ethernet1/0
      194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.8/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.16/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.24/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.32/29 [110/20] via 10.0.0.69, 00:44:47, Ethernet1/0
O E2     194.65.101.40/29 [110/20] via 10.0.0.69, 00:44:47, Ethernet1/0
O E2     194.65.101.48/28 [110/20] via 10.0.0.69, 00:44:47, Ethernet1/0
O E2     194.65.101.48/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.56/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.64/28 [110/20] via 10.0.0.69, 00:44:47, Ethernet1/0
O E2     194.65.101.64/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
O E2     194.65.101.72/29 [110/20] via 10.0.0.69, 00:45:36, Ethernet1/0
      194.65.103.0/29 is subnetted, 10 subnets
O        194.65.103.0 [110/30] via 10.0.0.69, 00:45:36, Ethernet1/0
O        194.65.103.8 [110/30] via 10.0.0.69, 00:45:36, Ethernet1/0
O        194.65.103.16 [110/40] via 10.0.0.69, 00:44:57, Ethernet1/0
O        194.65.103.24 [110/40] via 10.0.0.69, 00:44:57, Ethernet1/0
O        194.65.103.32 [110/50] via 10.0.0.69, 00:44:57, Ethernet1/0
O        194.65.103.40 [110/50] via 10.0.0.69, 00:44:57, Ethernet1/0
O        194.65.103.48 [110/50] via 10.0.0.69, 00:44:57, Ethernet1/0
O        194.65.103.56 [110/50] via 10.0.0.69, 00:44:57, Ethernet1/0
O        194.65.103.64 [110/40] via 10.0.0.69, 00:44:57, Ethernet1/0
O        194.65.103.72 [110/40] via 10.0.0.69, 00:44:57, Ethernet1/0
      194.65.104.0/24 is variably subnetted, 12 subnets, 2 masks
C        194.65.104.0/29 is directly connected, Ethernet0/2
L        194.65.104.1/32 is directly connected, Ethernet0/2
C        194.65.104.8/29 is directly connected, Ethernet0/3
L        194.65.104.9/32 is directly connected, Ethernet0/3
R        194.65.104.16/29 [120/1] via 10.0.6.2, 00:00:24, Ethernet0/0
R        194.65.104.24/29 [120/1] via 10.0.6.2, 00:00:24, Ethernet0/0
R        194.65.104.32/29 [120/2] via 10.0.6.2, 00:00:24, Ethernet0/0
R        194.65.104.40/29 [120/2] via 10.0.6.2, 00:00:24, Ethernet0/0
R        194.65.104.48/29 [120/1] via 10.0.6.17, 00:00:13, Ethernet0/1
R        194.65.104.56/29 [120/1] via 10.0.6.17, 00:00:13, Ethernet0/1
R        194.65.104.64/29 [120/2] via 10.0.6.17, 00:00:13, Ethernet0/1
R        194.65.104.72/29 [120/2] via 10.0.6.17, 00:00:13, Ethernet0/1
      194.65.105.0/24 is variably subnetted, 3 subnets, 2 masks
```

R-1:

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BG
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS lev
       ia - IS-IS inter area, * - candidate default, U - per-user static
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.0.73 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.0.0.73
      10.0.0.0/8 is variably subnetted, 37 subnets, 3 masks
O        10.0.0.36/30 [110/30] via 10.0.0.73, 00:44:48, Ethernet1/0
O        10.0.0.40/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O        10.0.0.44/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O        10.0.0.56/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O        10.0.0.60/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O        10.0.0.64/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O        10.0.0.68/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
C        10.0.0.72/30 is directly connected, Ethernet1/0
L        10.0.0.74/32 is directly connected, Ethernet1/0
O E2     10.0.3.0/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.0.3.4/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.0.3.8/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.0.3.12/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.0.3.16/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O        10.0.4.0/30 [110/30] via 10.0.0.73, 00:44:58, Ethernet1/0
O        10.0.4.4/30 [110/30] via 10.0.0.73, 00:45:08, Ethernet1/0
O        10.0.4.8/30 [110/40] via 10.0.0.73, 00:44:58, Ethernet1/0
O        10.0.4.12/30 [110/50] via 10.0.0.73, 00:44:58, Ethernet1/0
O        10.0.4.16/30 [110/40] via 10.0.0.73, 00:44:58, Ethernet1/0
O E2     10.0.5.0/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.0.5.4/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.0.5.8/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.0.6.0/30 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     10.0.6.4/30 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     10.0.6.8/30 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     10.0.6.12/30 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     10.0.6.16/30 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O        10.0.7.0/24 is a summary, 00:45:44, Null0
C        10.0.7.0/30 is directly connected, Ethernet0/0
L        10.0.7.1/32 is directly connected, Ethernet0/0
O        10.0.7.4/30 [110/20] via 10.0.7.2, 00:45:44, Ethernet0/0
C        10.0.7.8/30 is directly connected, Ethernet0/1
L        10.0.7.9/32 is directly connected, Ethernet0/1
O        10.0.7.12/30 [110/40] via 10.0.7.2, 00:45:39, Ethernet0/0
O        10.0.7.16/30 [110/30] via 10.0.7.2, 00:45:44, Ethernet0/0
O E2     10.3.3.12/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     10.3.3.16/30 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
     194.65.98.0/29 is subnetted, 2 subnets
O        194.65.98.56 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O        194.65.98.64 [110/30] via 10.0.0.73, 00:44:48, Ethernet1/0
     194.65.101.0/24 is variably subnetted, 12 subnets, 2 masks
O E2     194.65.101.0/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     194.65.101.8/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     194.65.101.16/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     194.65.101.24/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     194.65.101.32/29 [110/20] via 10.0.0.73, 00:44:48, Ethernet1/0
```

```
O E2     194.65.101.40/29 [110/20] via 10.0.0.73, 00:44:48, Ethernet1/0
O E2     194.65.101.48/28 [110/20] via 10.0.0.73, 00:44:48, Ethernet1/0
O E2     194.65.101.48/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     194.65.101.56/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     194.65.101.64/28 [110/20] via 10.0.0.73, 00:44:48, Ethernet1/0
O E2     194.65.101.64/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
O E2     194.65.101.72/29 [110/20] via 10.0.0.73, 00:45:44, Ethernet1/0
     194.65.103.0/29 is subnetted, 10 subnets
O        194.65.103.0 [110/30] via 10.0.0.73, 00:45:44, Ethernet1/0
O        194.65.103.8 [110/30] via 10.0.0.73, 00:45:44, Ethernet1/0
O        194.65.103.16 [110/40] via 10.0.0.73, 00:44:58, Ethernet1/0
O        194.65.103.24 [110/40] via 10.0.0.73, 00:44:58, Ethernet1/0
O        194.65.103.32 [110/50] via 10.0.0.73, 00:44:58, Ethernet1/0
O        194.65.103.40 [110/50] via 10.0.0.73, 00:44:58, Ethernet1/0
O        194.65.103.48 [110/50] via 10.0.0.73, 00:44:58, Ethernet1/0
O        194.65.103.56 [110/50] via 10.0.0.73, 00:44:58, Ethernet1/0
O        194.65.103.64 [110/40] via 10.0.0.73, 00:44:58, Ethernet1/0
O        194.65.103.72 [110/40] via 10.0.0.73, 00:44:58, Ethernet1/0
     194.65.104.0/29 is subnetted, 10 subnets
O        194.65.104.0 [110/30] via 10.0.0.73, 00:45:39, Ethernet1/0
O        194.65.104.8 [110/30] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     194.65.104.16 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     194.65.104.24 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     194.65.104.32 [110/20] via 10.0.0.73, 00:45:27, Ethernet1/0
O E2     194.65.104.40 [110/20] via 10.0.0.73, 00:45:27, Ethernet1/0
O E2     194.65.104.48 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     194.65.104.56 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     194.65.104.64 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
O E2     194.65.104.72 [110/20] via 10.0.0.73, 00:45:39, Ethernet1/0
     194.65.105.0/24 is variably subnetted, 13 subnets, 3 masks
O        194.65.105.0/24 is a summary, 00:45:44, Null0
C        194.65.105.0/29 is directly connected, Ethernet0/2
L        194.65.105.1/32 is directly connected, Ethernet0/2
C        194.65.105.8/29 is directly connected, Ethernet0/3
L        194.65.105.9/32 is directly connected, Ethernet0/3
O        194.65.105.16/29 [110/20] via 10.0.7.2, 00:45:44, Ethernet0/0
O        194.65.105.24/29 [110/20] via 10.0.7.2, 00:45:44, Ethernet0/0
O        194.65.105.32/29 [110/30] via 10.0.7.2, 00:45:44, Ethernet0/0
O        194.65.105.40/29 [110/30] via 10.0.7.2, 00:45:44, Ethernet0/0
O        194.65.105.48/29 [110/50] via 10.0.7.2, 00:45:39, Ethernet0/0
O        194.65.105.56/29 [110/50] via 10.0.7.2, 00:45:39, Ethernet0/0
O        194.65.105.64/29 [110/40] via 10.0.7.2, 00:45:39, Ethernet0/0
O        194.65.105.72/29 [110/40] via 10.0.7.2, 00:45:39, Ethernet0/0
```

- *Verificação de conetividades:*

```
EAC8#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/8/17 ms
```

```
HP1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/4 ms
```

```
BV1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/8/19 ms
```

```
SO1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/8/23 ms
```

```
SWC1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/8/18 ms
```

```
CC1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/5/10 ms
```

```
ADL1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/9/19 ms
```

```
R1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/8/18 ms
```